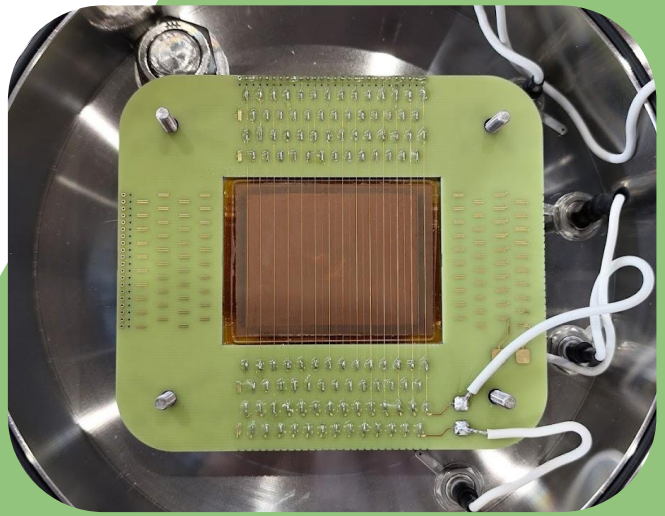
The MacroStrip Proportional Chamber

A new tracking detector for large area applications



M. Giovannetti

Laboratori Nazionali di Frascati - INFN

Grant Giovani CSN5, Dec. 2nd 2025

The MSPC concept

THE GOAL:

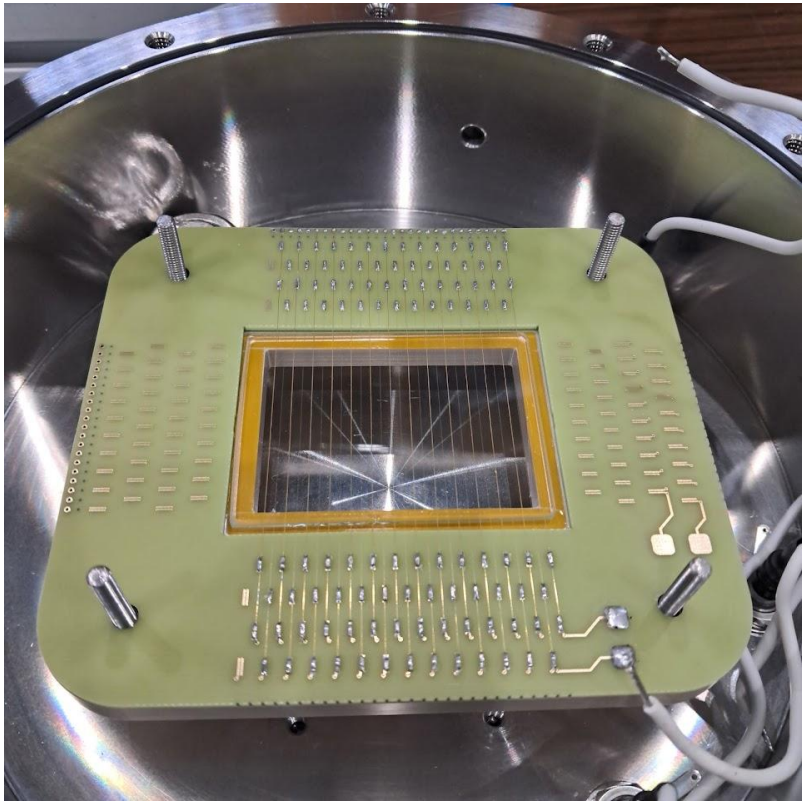
- Develop a new **gaseous particle detector** for **mid rate** applications: **O(200 kHz/cm^2)**
- **Space resolution down to 100 µm**
- **Time resolution down to 6 ns**
- **Easier** construction w.r.t. MWPC
  - reduced assembly person-power

HOW:

- Using **photolithographic techniques** from **MicroPattern Gaseous Detetors**

WHY:

- for **all low-mid rate applications where MWPC would still be valid** but no longer feasible due to loss in expertise or high cost due to time-consuming procedure



Spillovers

NOW

Tracking applications below 200 kHz/cm^2 are still common

CMS, ATLAS, and LHCb **still use** MWPC, DT, CSC, sTGCs…

Beam monitoring at PS and SPS beam lines is performed with **Wire** detectors: **CERN need to build new MWPCs**

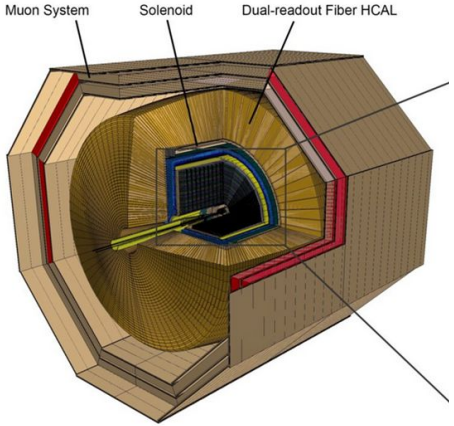
The MSPC is suitable for all these application and more: **it will be the upgrade of MWPCs**

Beyond HEP the main advantage is the coverage of **large area with low cost**:

muon radiography/tomography (volcanoes, civil structures)
neutron detection - **homeland security**
neutron detection - **position sensitive detectors**

Future e^+e^- colliders

The **MSPC** is a good candidate for the **IDEA Muon system**



X-Y readout w/ space resolution <**200 µm**
Total surface: **1500m^2, ∼6000 detector**
Particle rate: **≪1 kHz/cm^2**

The candidate

Competence and responsibility:

- R&D on μ-RWELL/MPGD since **2017 (MsC, PhD, Postdoc)**
  - High rate layouts, large size: design, assembly, operation
  - **TT expert at ELTOS S.p.A.** (IT)
- R&D on sRPC since 2020
- Test beam and data taking responsibilities
- Cooperation experience in **international working groups (RD51, DRD1)**

Academic and Working Experience:

- **μ-RWELL** for thermal neutron detection - CERN ATTRACT
- **sRPC & μ-RWELL** for thermal neutron detection - uRANIA (**CSN-5**)
- **sRPC** for thermal neutron detection - ATHENA (**CSN-5**)
- **μ-RWELL** for LHCb Phase II Upgrade (**CSN1**)
- **μ-RWELL** for FCC-IDEA muon (**CSN1**)
- **μ-RWELL** TT - INFN-AIDAinnova (**CSN1**)
- **cylindrical μ-RWELL** development - INFN-EURIZON (**CSN1**)





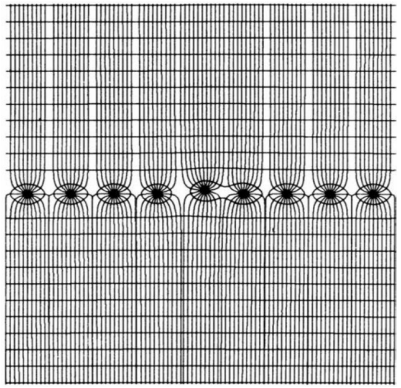
State of the art: MWPC



Charpak, 1968

The MWPC: a revolution in HEP experiments

- G. Charpak: MWPC invented in 1968 (Nobel 1992)
- **better rate capability w.r.t. detectors in the ‘60s**
  (a necessity while increasing experiments luminosity)
  - **still limited at O(200 kHz/cm^2)**
- space resolution 50-100 μm
- time resolution 5-10 ns
- efficiency close to 100% for m.i.p.



Manufacturing problems:

- a **very high personpower** needed for each detector
- **very few production places** still available in the world

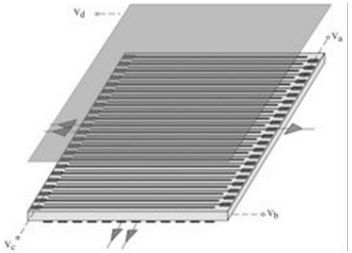
E.g.: **LHCb U2 new MWPC** @ LNF

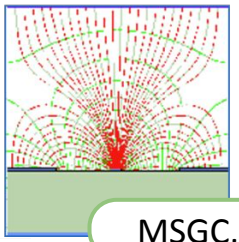

LNF, 2025

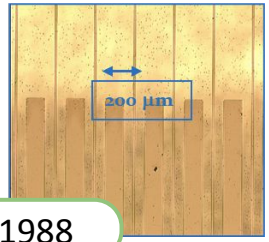
State of the art: MPGD

MPGDs: another revolution in HEP experiments

- A. Oed: MSGC invented in 1988 (no Nobel)
- **anode-cathode distance O(100 µm)**
  - **rate capability up to O(10 MHz/cm^2)**
  - **more prone to destructive sparks w.r.t. MWPC**
- industrial photolithographic techniques for fast and large size development



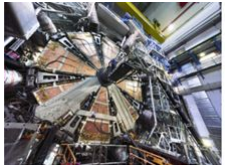


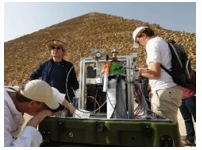


MSGC, 1988

COMPASS (Saclay) ATLAS (LNF) Giza (CEA)



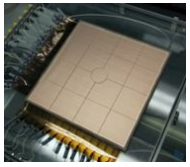




MM, 1996

COMPASS (CERN) LHCb (LNF) KLOE-2 (LNF)





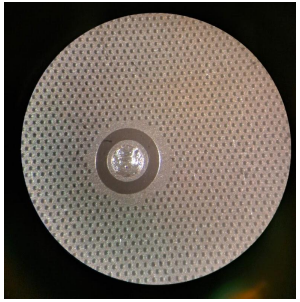




GEM, 1997

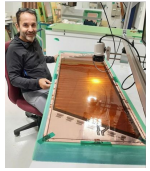


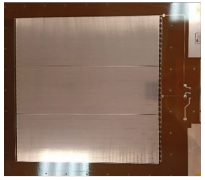
LNF, 2025

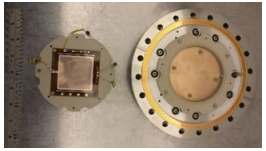


CLAS12 (JLAB) n_TOF (CERN) UKRI (UK)









µ-RWELL, 2014

2026: can we get the best of both worlds?



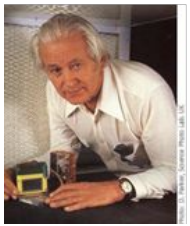
MWPC, 2025

Multi**W**ire **P**roportional **C**hambers

- **handmade**
- 600 wires - 2mm pitch
  - wiring/tensioning
  - gluing (24h curing)
  - soldering
- **4 detectors / 7 working days /4 technicians**









MPGD, 2025

Micro**P**attern
Gaseous **D**etectors

- **industrial process**

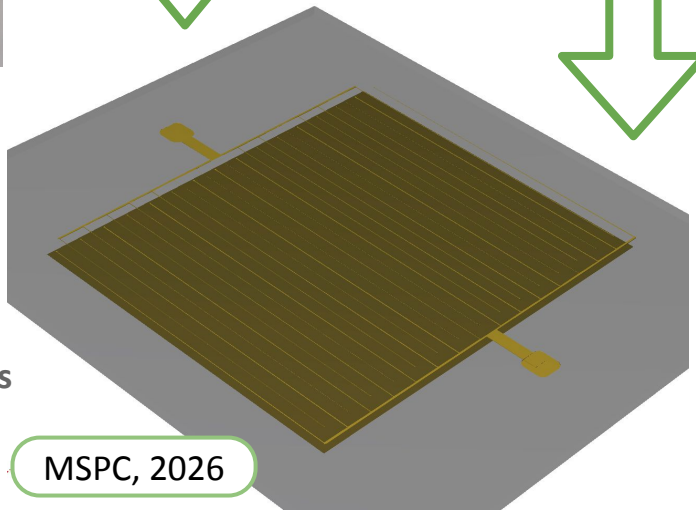
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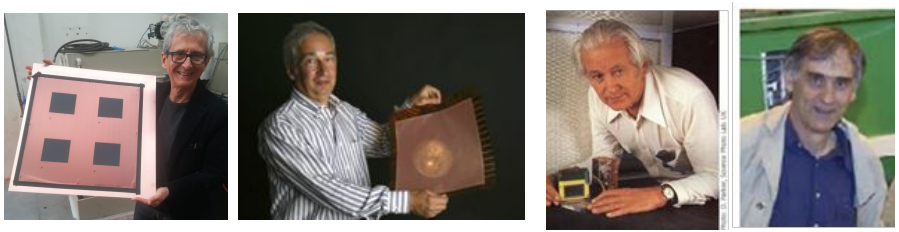
MWPC, 2025

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MSPC, 2026



MPGD, 2025

Micro**P**attern **G**aseous **D**etectors

- **industrial process**

The **M**acro**S**trip **P**roportional **C**hamber

- **industrial process**
  - amplification stage “ready made”
  - only to be closed w/ cathode
- **4 detectors / 1 working day /2 technicians**

The MSPC - MacroStrip Proportional Counter

A “wireless” MWPC:

- **Field** (**100µm** width) and **Sense** (**30µm** width) metallic strips
- resistive **glass substrate** + backplane
- Field-Sense distance: **2mm**





Advantages over MWPC:

- **easier to build** (reduced personpower: photolithographic process)

Advantages over MPGD:

- **good spark suppression** (similar to MWPC)

The MSPC - MacroStrip Proportional Counter

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Advantages over MWPC:

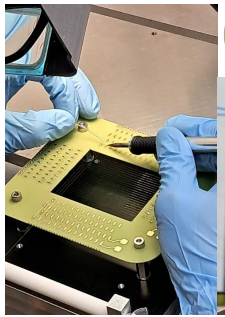
- **easier to build** (reduced personpower: photolithographic process)

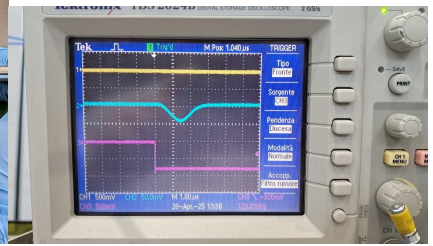
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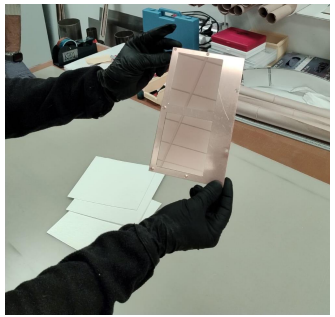
WHY IT WILL WORK:

Proof of operation, LNF

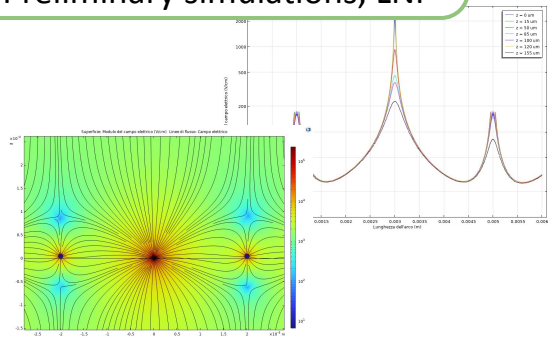




Material study, CERN



Preliminary simulations, LNF

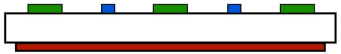


Proof of Operation @ LNF

Is glass resistivity compatible with the detector operation?



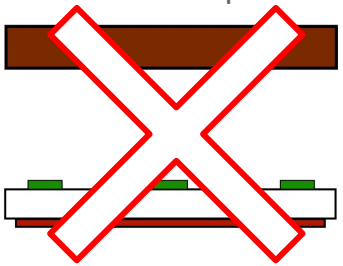
CATHODE



SENSE STRIP
FIELD STRIP
GLASS
BACK-PLANE

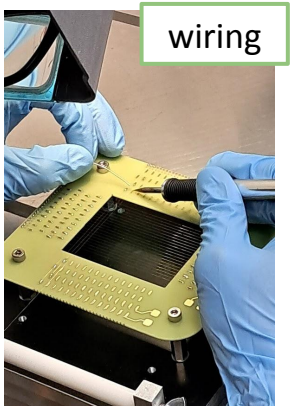
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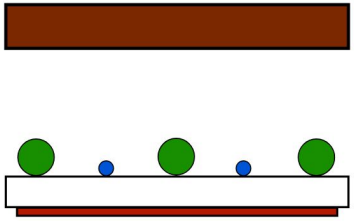


CATHODE

SENSE STRIP
FIELD STRIP
GLASS
BACK-PLANE



Using 100um/30um **wires** on glass, to **simulate the metallic strips**

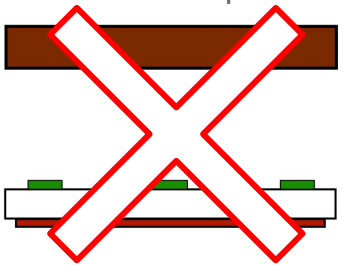


CATHODE

SENSE WIRES
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BACK-PLANE

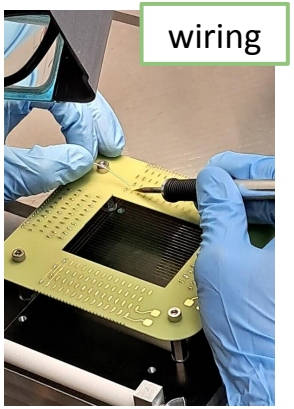
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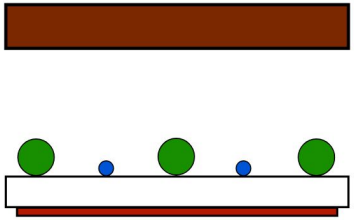
CATHODE

SENSE STRIP
FIELD STRIP
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BACK-PLANE



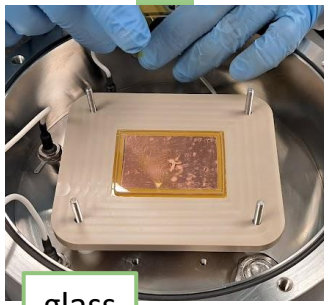
wiring

Using 100um/30um **wires** on glass, to **simulate the metallic strips**



CATHODE

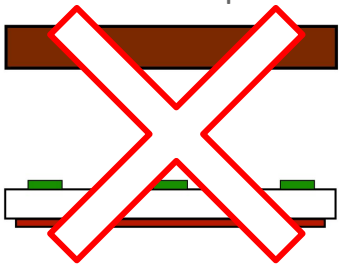
SENSE WIRES
FIELD WIRES
GLASS
BACK-PLANE



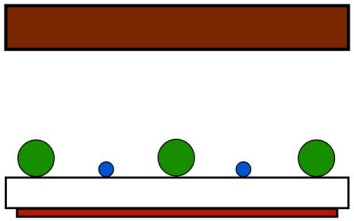
glass

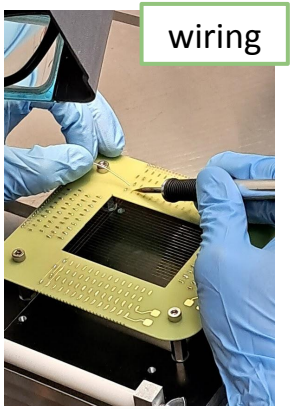
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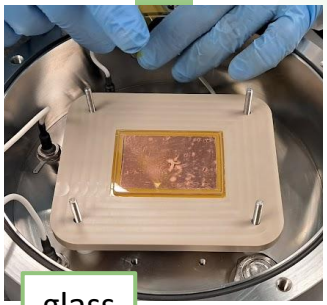
Is glass resistivity compatible with the detector operation?

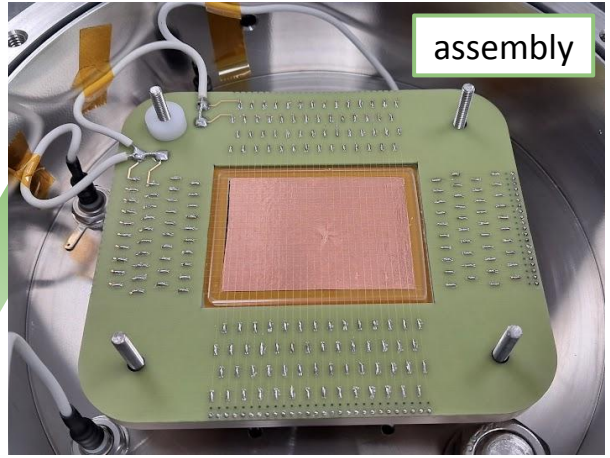


Using 100um/30um **wires** on glass, to **simulate the metallic strips**



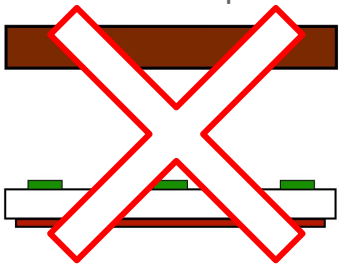




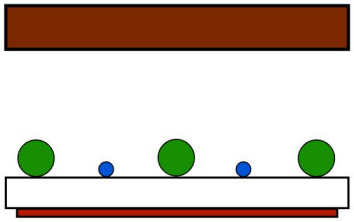


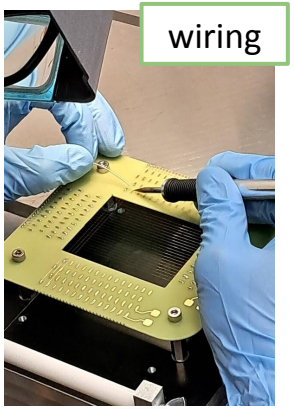
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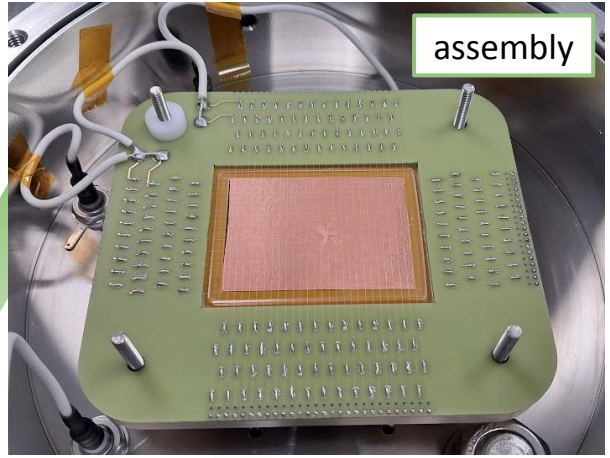
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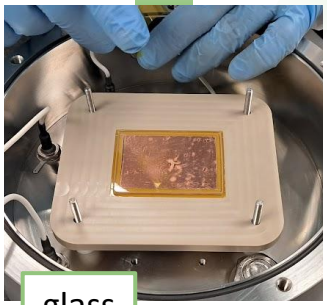


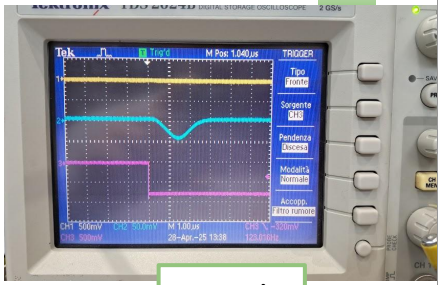
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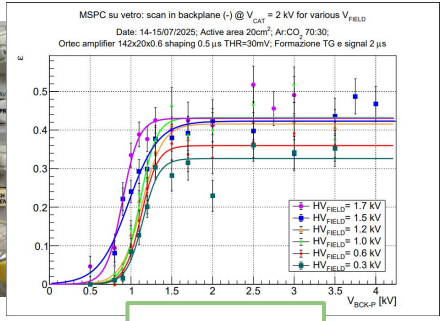



wiring


assembly


glass


signals



it works!

Material study @ CERN



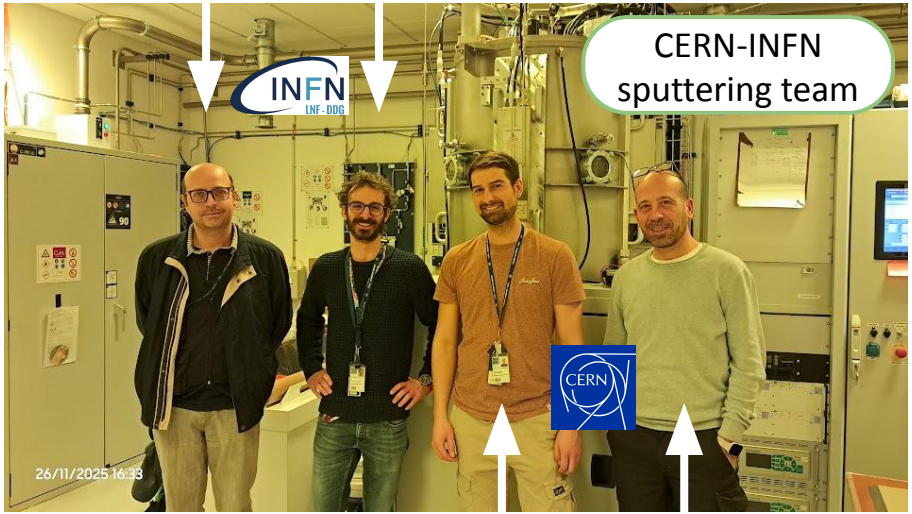




Standard deposition:
2 µm Cu (strips)
30 nm Ti
1.8 mm **glass**
30 nm Ti
2 µm Cu (backplane)

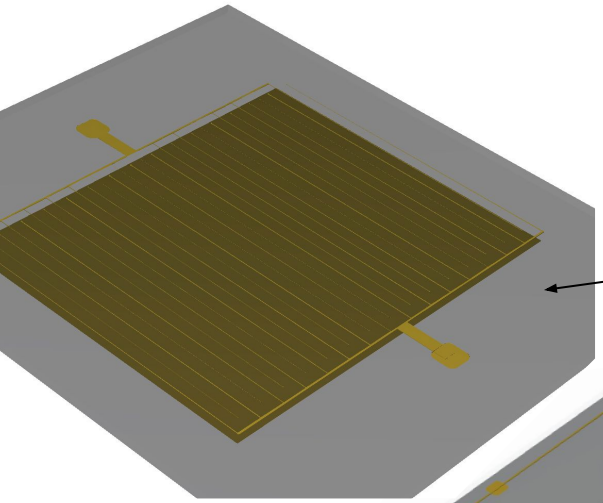
The “**CERN-INFN** magnetron **sputtering machine**”
- (60x170cm^2) flexible and **(20x50cm^2) rigid substrate**
- deposition: C, **Cu**, Ti, Al

Gianfranco (tech. manager) and Gianluca are involved in the MSPC project, this will ensure major control over the sputtering process.



Karl and Hector will be also responsible at CERN of the chemical etching of the amplification stage of the MSPC.

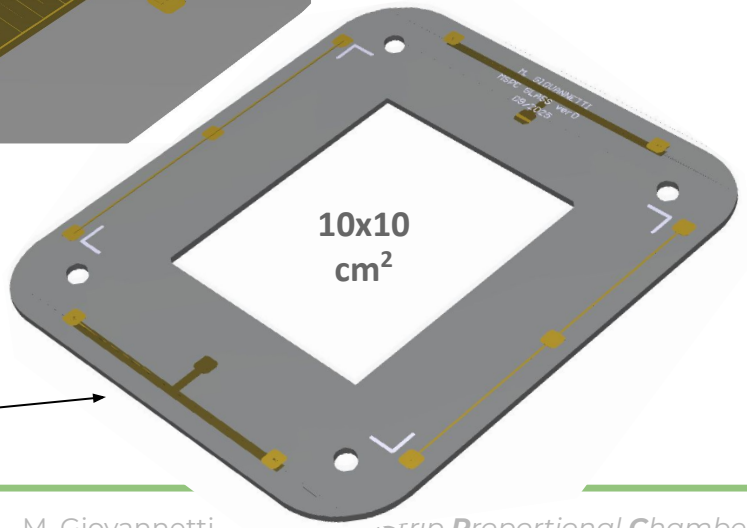
The Prototype - Design



SENSE strip
(30 µm)

FIELD strip
(100 µm)

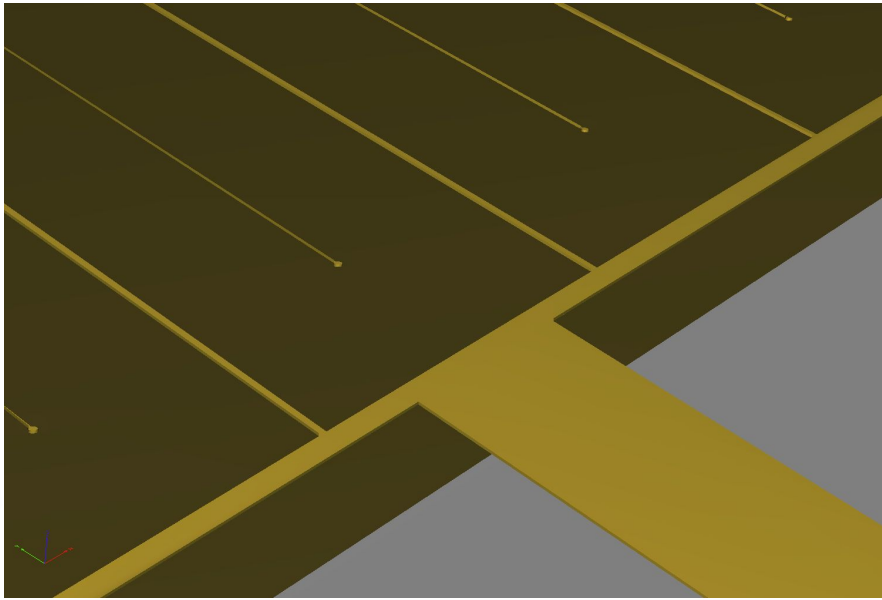
GLASS



support
PCB



SENSE STRIP
FIELD STRIP
GLASS
BACK-PLANE



Work Packages

Goals of the MSPC project is

1st YEAR - M2 ➔ to design, build and operate the **1st prototype of MSPC** (10x10cm^2)

2nd YEAR - D1 ➔ to build a **real size MSPC detector** (20x50cm^2)

Work Packages

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WP1 - Mechanics, design and construction

M1 ➔ T1.1 - Amplification stage design of small area

T1.2 - Mechanical design of vessel of small area

T1.3 - HV distribution design

M2 ➔ T1.4 - Base material and amplification stage production of small area

D1 ➔ T1.5 - Assembly in clean room

T1.6 - Amplification stage design of large area

T1.7 - Mechanical design of vessel of large area

T1.8 - Base material and amplification stage production of large area

Work Packages

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T1.6 - Amplification stage design of large area
T1.7 - Mechanical design of vessel of large area
T1.8 - Base material and amplification stage production of large area



WP2 – Detector simulation and reconstruction

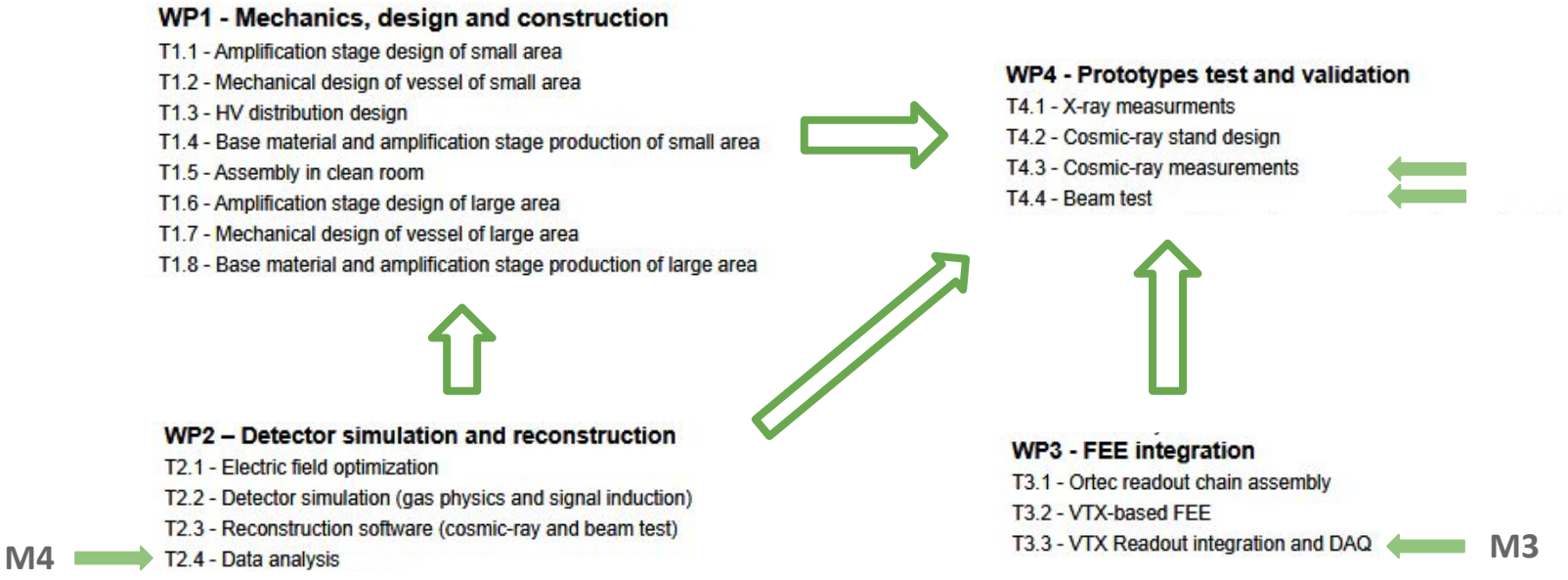
T2.1 - Electric field optimization
T2.2 - Detector simulation (gas physics and signal induction)
T2.3 - Reconstruction software (cosmic-ray and beam test)
T2.4 - Data analysis

Work Packages

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Timeline

WP1 - Mechanics, design and construction
T1.1 - Amplification stage design of small area
T1.2 - Mechanical design of vessel of small area
T1.3 - HV distribution design
T1.4 - Base material and amplification stage production of small area
T1.5 - Assembly in clean room
T1.6 - Amplification stage design of large area
T1.7 - Mechanical design of vessel of large area
T1.8 - Base material and amplification stage production of large area
WP2 – Detector simulation and reconstruction
T2.1 - Electric field optimization
T2.2 - Detector simulation (gas physics and signal induction)
T2.3 - Reconstruction software (cosmic-ray and beam test)
T2.4 - Data analysis
WP3 - FEE integration
T3.1 - Ortec readout chain assembly
T3.2 - VTX-based FEE
T3.3 - VTX Readout integration and DAQ
WP4 - Prototypes test and validation
T4.1 - X-ray measurments
T4.2 - Cosmic-ray stand design
T4.3 - Cosmic-ray measurements
T4.4 - Beam test
MILESTONE - Beam test result presented in conference/workshop



M1	Amplification stage design for small area detectors	m4
M2	Production of 16 small area amplification stages	m9
M3	Development of the VTX FEE board and R/O chain	m12
M4	Software for characterization of small area detectors	m16
M5	Beam test results presented in conference/workshop	m24
D1	Production of a large area detector	m19

10x10cm^2 prototypes

Total 1st year 73.9 k€

20x50cm^2 detector

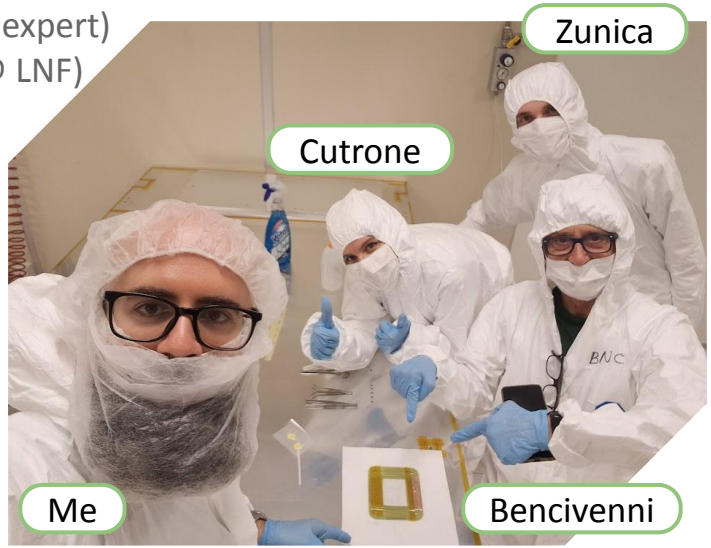
Total 2nd year 61.9 k€

Team Composition

- *Main workforce*
  1.00 - **M. Giovannetti** (LNF - PI, AdR sr.)
  1.00 - **G. Zunica** (LNF - AdR sr. Detector Development service @ LNF)
  ……. - **G. Cutrone** (LNF summer student during 08/25, now Sapienza student, MsC thesis on MSPC from 06/26)
- *Practical support:*
  0.20 - **M. Poli Lener** (LNF - gaseous detector expert)
  0.20 - **G. Morello** (LNF - Sputtering machine tech. manager, mechanical expert)
  0.05 - **M. Beretta** (LNF - unit head of Electronic & Automation service @ LNF)

- *Scientific advisors:*
  0.30 - **G. Bencivenni** (LNF - *retiring on 04/26* - gaseous detector expert, head of Detector Development service @ LNF)
  0.20 - **P. De Simone** (LNF - *retiring on 09/26* - simulations and analysis)
  0.15 - **G. Felici** (LNF - *retired in '24* - FEE expert)

  FTE total = 3.10 + MsC student



Synergies

The **CERN-INFN Magnetron Sputtering machine** (CID) is a key asset for this project. Exploring different MSPC stacking the CID team expertise will increase.

The **CERN-MPT-Workshop** is leader in developing new MPGD technologies (MSGC, GEM, MicroMegas, μ-RWELL) and is actively involved in each INFN activity regarding MPGDs.

Collaboration with existing industries: **ELTOS S.p.A.** is a consolidated INFN and CERN partner for developing PCBs for electronics and Detectors. ELTOS is also engaged in the INFN EU project **AIDAinnova/OTELLO**.

MPT Workshop at LNF - INFN

Project of the "**Detectors Development and Experimental Activities Support**" service, *commissioning from 2027-Q3*.
Independent research unit and service for external (INFN) users.

Goals:

- **Develop new MPGDs** (prototypes)
- Foster technology development inside LNF through the use of industrial techniques
- Be part of the international link of MPGD developer labs (CERN, Saclay, Bonn, JLab, USTC-Hefei)

Technologies required:

- **Material coating** (DLC machine)
  - Diamond-Like-Carbon (DLC), B_4C, Cu, Al, Cr, Ti, …
- **Material Patterning**
  - **Dry film deposition/patterning,** DLC patterning (jet-scrubbing, brushing machine)
- **Chemical Etching**
  - **Controlled removal of Cu Cr, Ti**

The MSPC could be manufactured at the LNF MPT Workshop

Summary

The MSPC concept **merges MWPC** robustness and simplicity **with MPGD** industrial photolithographic fabrication, **enabling large-area tracking** with O(100 μm) spatial resolution, O(10 ns) timing, O(10^5 Hz/cm²) rate capability and **a great reduction in assembly person-power.**

In the next two years I will lead the simulation, design, construction and test of prototypes, establishing a validated roadmap:
from proof-of-concept to real detector.

To carry on this project I will exploit consolidated **INFN–LNF and CERN infrastructures**, helped by a **team** with proven **expertise in MWPC, GEM and μ-RWELL detectors** and **an industrial partner** already proficient in MPGD photolithography.

With this grant, **MSPC will become a competitive option** for HEP and non-HEP applications, providing a cost-effective, **industrially scalable alternative to traditional wire chambers**.

SPARE

Spare - Budget and Costs

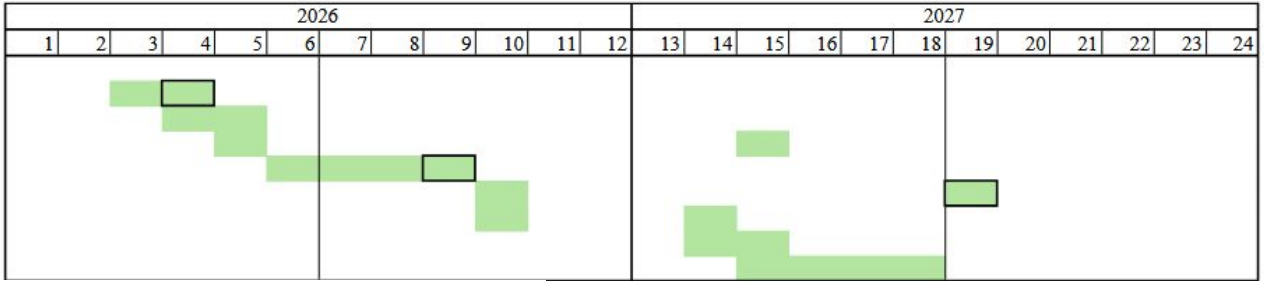
Item	Quantity	Unitary Cost	Total Cost	WP	Quotation
Consumable					
Mechanical design of vessel for small area	4	0,45	**1,8**	1.2	
HV distribution for small area	4	0,45	**1,8**	1.3	
Base material and amplification stage production for small area	16	1,29	**20,7**	1.4	CERN_Small_Area.pdf
Assembly in clean room	1	1,90	**1,9**	1.5	
VTX-based FFE	6	4,17	**25,0**	3.2	VTX_FEE.pdf
Gas Bottles	4	1,00	**4,0**	4.3	Rivoira_gas.pdf
Cosmic-ray stand	1	1,90	**1,9**	4.2	
Inventory					
PC purchase for simulation	1	1,45	**1,5**	2.2	PC_invoice.pdf
		First year	**58,5**		
		IVA	**71,4**		
Mission					
Contat with production sites	1	2,5	**2,5**	1.4	
		Total first year	**73,9**		

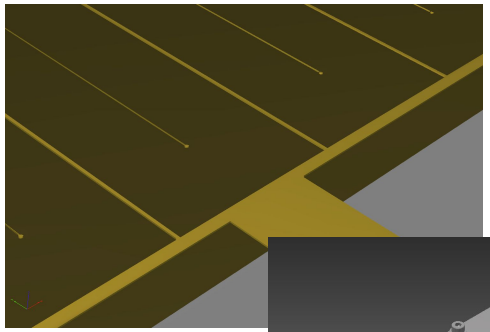
Consumable					
Mechanical design of vessel of large area	2	0,90	**1,8**	1.7	
HV distribution design for large area	2	0,90	**1,8**	1.3	
Base material and amplification stage production of large area	4	6,27	**25,1**	1.6	CERN_Large_Area.pdf
Inventory					
HV Board Caen	1	8	**8**	4.3	CAEN_invoice.pdf
		Second year	**36,7**		
		IVA	**44,8**		
Mission					
Mission for conference	1	2,5	**2,5**	4.4	
Mission for TB crew	4	3	**12**	4.4	
Mission for TB transport	2	1,3	**2,6**	4.4	
		Total second year	**61,9**		

WP1: Mechanics, Design & Construction

WP1 - Mechanics, design and construction

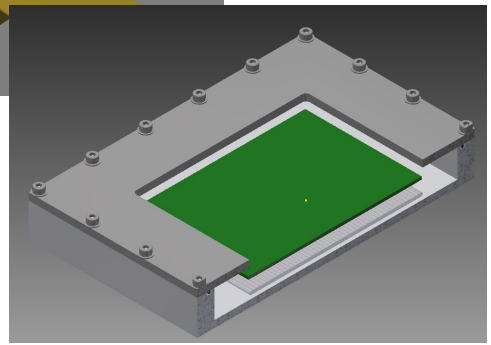
T1.1 - Amplification stage design of small area
T1.2 - Mechanical design of vessel of small area
T1.3 - HV distribution design
T1.4 - Base material and amplification stage production of small area
T1.5 - Assembly in clean room
T1.6 - Amplification stage design of large area
T1.7 - Mechanical design of vessel of large area
T1.8 - Base material and amplification stage production of large area







SENSE STRIP
FIELD STRIP
GLASS
BACK-PLANE



M. Giovannetti *+ G. Bencivenni*

- Amplification stage design (T1.1, T1.6)

G. Zunica + G. Morello

- Amplification stage production (T1.4, T1.8)

G. Morello

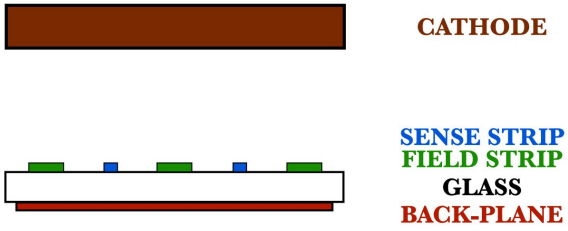
- Mechanical Design (T1.2, T1.3,T1.7)

full team

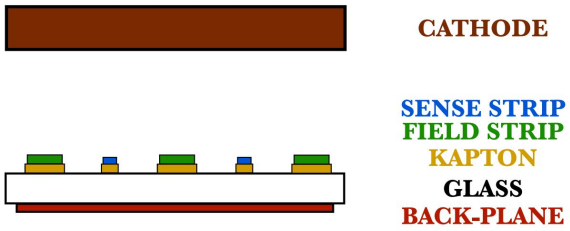
- Clean Room Assembly (T1.5)

WP1 - Possible MSPC configurations

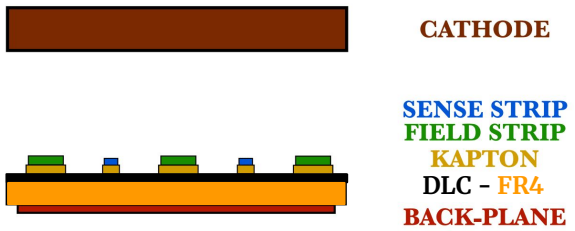
MSPC-2a sketch



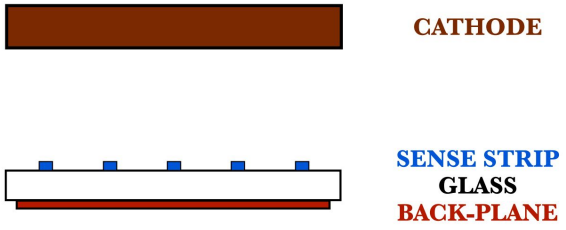
MSPC-2ak sketch



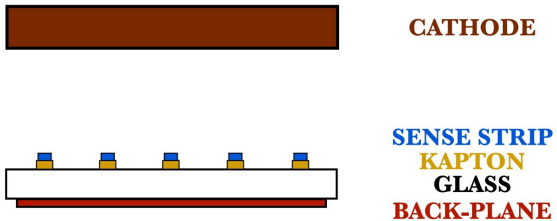
MSPC-2ak with DLC



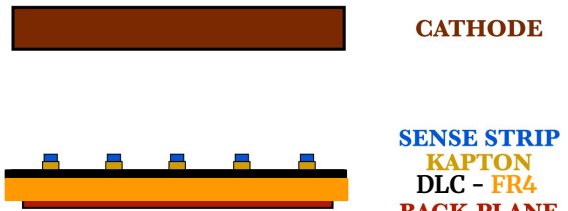
MSPC-2u sketch



MSPC-2uk sketch



MSPC-2uk with DLC



WP2: Detector Simulation & Reconstruction

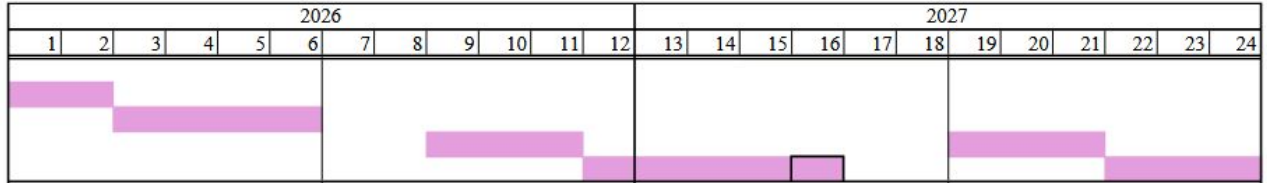
WP2 – Detector simulation and reconstruction

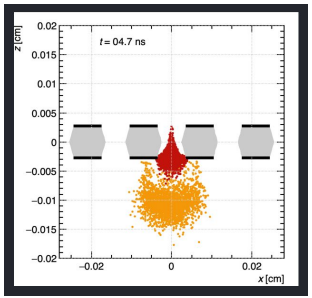
T2.1 - Electric field optimization

T2.2 - Detector simulation (gas physics and signal induction)

T2.3 - Reconstruction software (cosmic-ray and beam test)

T2.4 - Data analysis





G. Zunica

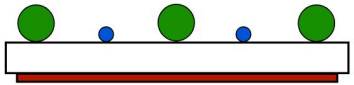
- Detector Simulation (T2.1, T2.2)
- Reconstruction Software (T2.3)

& w/ M. Giovannetti

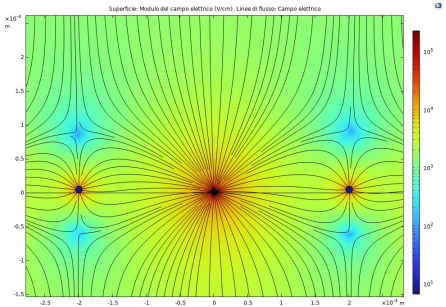
- Data analysis (T2.4)

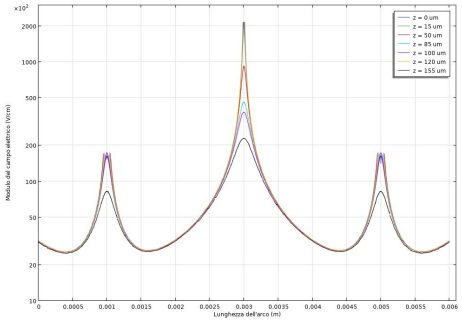


CATHODE



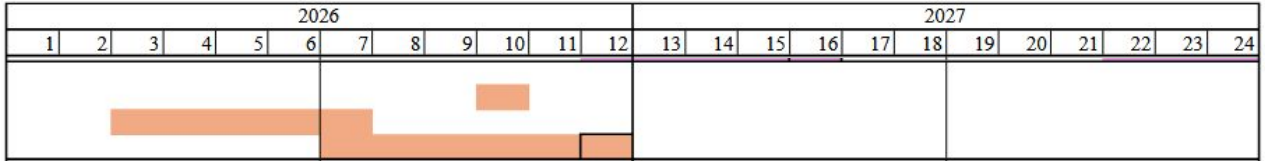
SENSE WIRES
FIELD WIRES
GLASS
BACK-PLANE



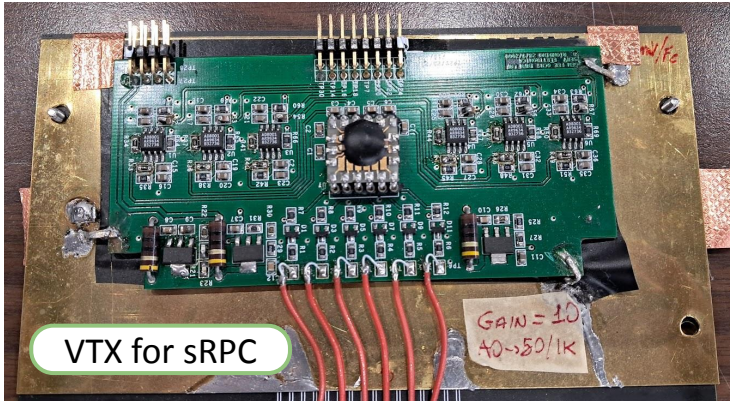


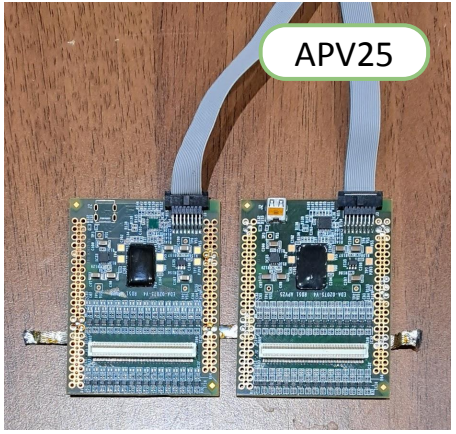
WP3: FEE Integration

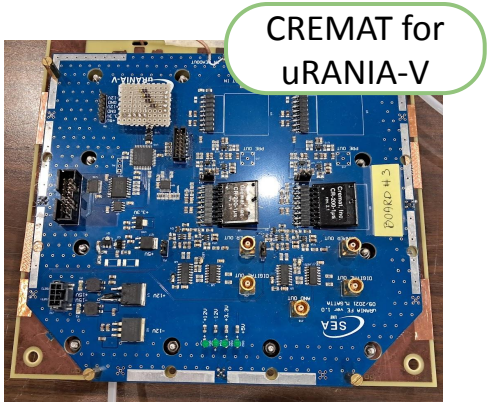
WP3 - FEE integration
T3.1 - Ortec readout chain assembly
T3.2 - VTX-based FEE
T3.3 - VTX Readout integration and DAQ



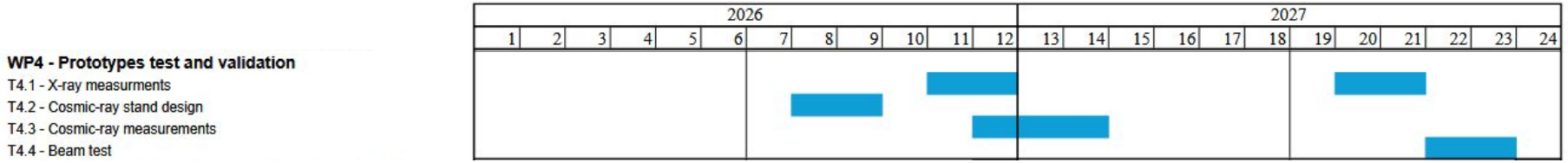
M. Giovannetti + G. Cutrone
Ortec readout chain (T3.1)
M. Beretta + *G. Felici*
VTX FEE development and integration (T3.2, T3.3)







WP4: Prototypes test and validation



G. Morello
Cosmic-ray stand design (T4.2)
M. Giovannetti + G. Cutrone
Data taking (T4.1, T4.3)
Full team
Beam test (T4.4)

Cosmic-ray stand installed at LNF-DDG laboratory (T4.2)

- first measurements will be conducted using the ORTEC pre-amplifier connected to the OR-ed strips, to quickly helps to find the optimal layout in terms of efficiency and stability
- further measurement will be carried out with the VTX-chain system (space and time resolution: T4.3)

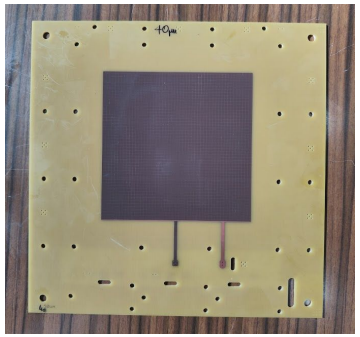
The same stand can be adapted for a beam test (T4.4)

- performed during the second year at the πM1 beam (PSI), in parasitic mode to LHCb setup (300 MeV/c muon beam)

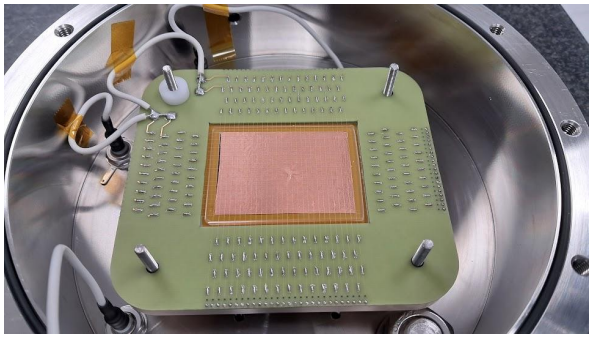
Risk Assessment

	Event	Risk	Mitigation
WP1	Issues with amplification stage production on glass (e.g.: delamination)	Inability to produce detectors at CERN	This process is standard in other industries, such as [21,22,23], using a different techniques (e.g. laser etching). However, the offer could not be available before September.
	Issues with amplification stage production on glass (e.g.: delamination)	Inability to use glass as substrate	Use of flexible substrate (Kapton) glued over FR4 (standard MPGD technology). Kapton can be fully etched in order to create other microstructures.
	Delay of the amplification stage for small area detector	Delay of the production of large area detectors and their beam test characterization	Beam test will be performed with the small area detectors
WP2	Material procurement for VTX FEE is delayed over the first year	Impossible to measure space resolution of small area prototypes with cosmic-ray	Cosmic-ray measurements will be performed with the Ortec 142 amplifier and large area detector design will be accordingly chosen
	Material procurement for VTX FEE is delayed over the second year	Impossible to use the VTX FEE for the beam test	Beam test will be performed with APV25 FEE borrowed from other INFN groups
	The MSPC gas gain is too low and the VTX gain is not enough to amplify the signals, resulting in a low S/N	Impossibility to measure the performance with the VTX FEE	The measurement can be done using Ortec 142 amplifier and APV25 FEE
WP4	Unavailability of beam time at PSI	Inability to perform the beam test and measure large area detector performance	Beam time request at common Beam Test Facility at LNF (or DESY)

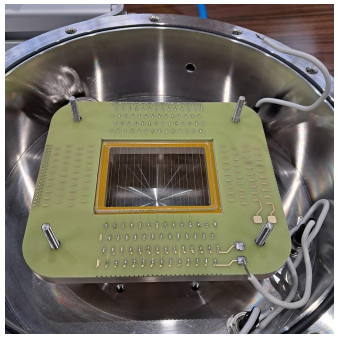
Proof of operation - Alternative configurations tested



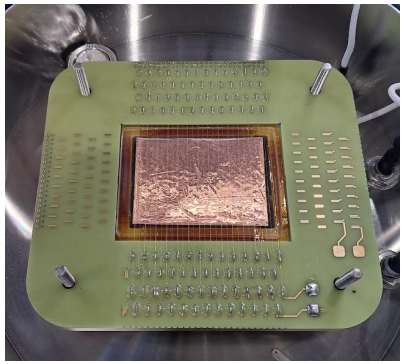
All FR4



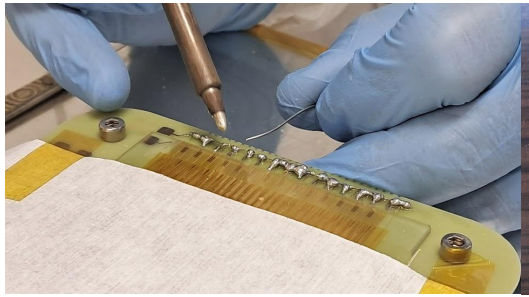
ALL Sense wires + backplane

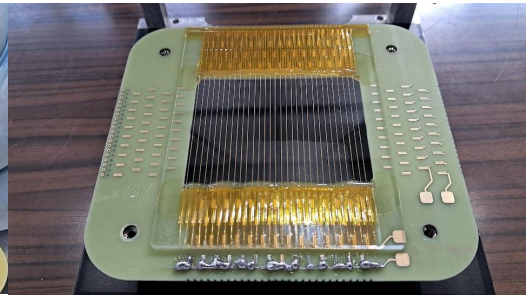


Cathodic-Sense wires but NO backplane



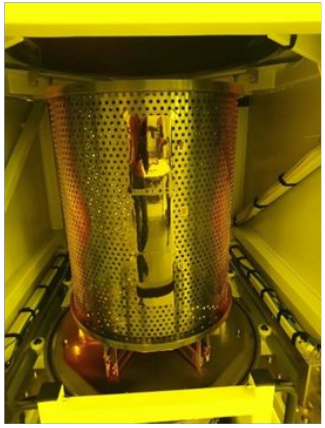
Cathodic-Sense wires + insulated backplane





Attempt to ensure **maximum adhesion between wires and glass**.
Even a tiny gap would create a zone with an electric field about **1MV/cm** !!
This would increase the dark counts (**with strips this problem will NOT be present**)

The CERN-INFN DLC Sputtering Machine (CID)



The **CID** (CERN-INFN-DLC) sputtering machine, a **joint project between CERN and INFN**, is used for preparing the **base material of the detector**. The potential of the sputtering machine is:

- **Flexible substrates** up to **1.7×0.6m^2**
- **Rigid substrates** up to **0.2×0.6m^2**

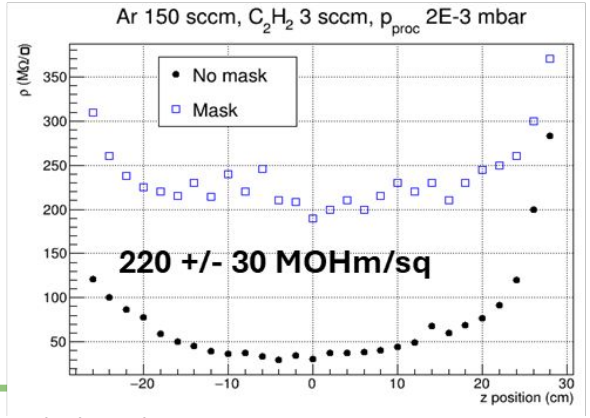
In **2023-24,** the activity on CID focused on the **tuning** of the **machine on small foils: good** results in terms of **reproducibility and uniformity.**

In **2025**

- **0.8×0.6m^2** foils have been sputtered with DLC or DLC+Cu
- **5x5cm^2** glass samples have been sputtered with DLC, Ti+Cu, Al, B_4C









The μ-RWELL TT

Step 0 – Detector **PCB design** @ LNF + CERN-MPT

Step 1 – CERN_INFN **DLC sputtering machine** @ CERN (+INFN)
- In operation since Nov. 2022
- Production by LNF-INFN crew

Step 2 – **Producing readout PCB** by ELTOS
- pad/strip readout

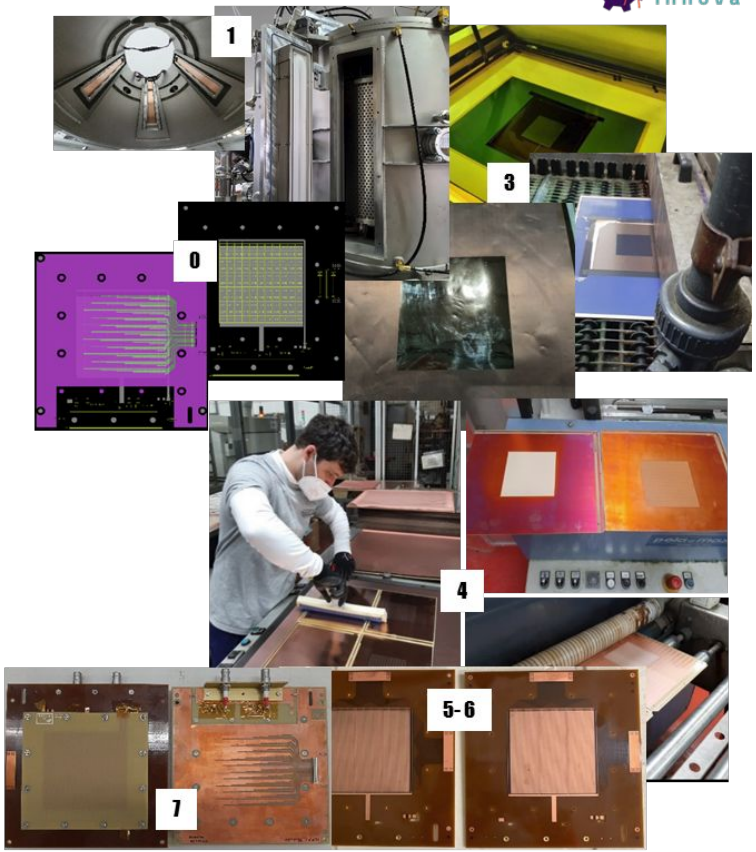
Step 3 – **DLC patterning** by ELTOS
- photo-resist → patterning with BRUSHING-machine

Step 4 – **DLC foil gluing** on PCB by ELTOS
- Large press available, up to 16 PCBs workable at the same time

Step 5 – **Ground network connections** creation by CERN
- **PEP** layout: Cu **P**atterning → PI **E**tching → Cu **P**lating

Step 6 – **Amplification stage** patterning by CERN
- Cu amplification holes image and HV connections by Cu etching
- PI etching → plating → amplification-holes

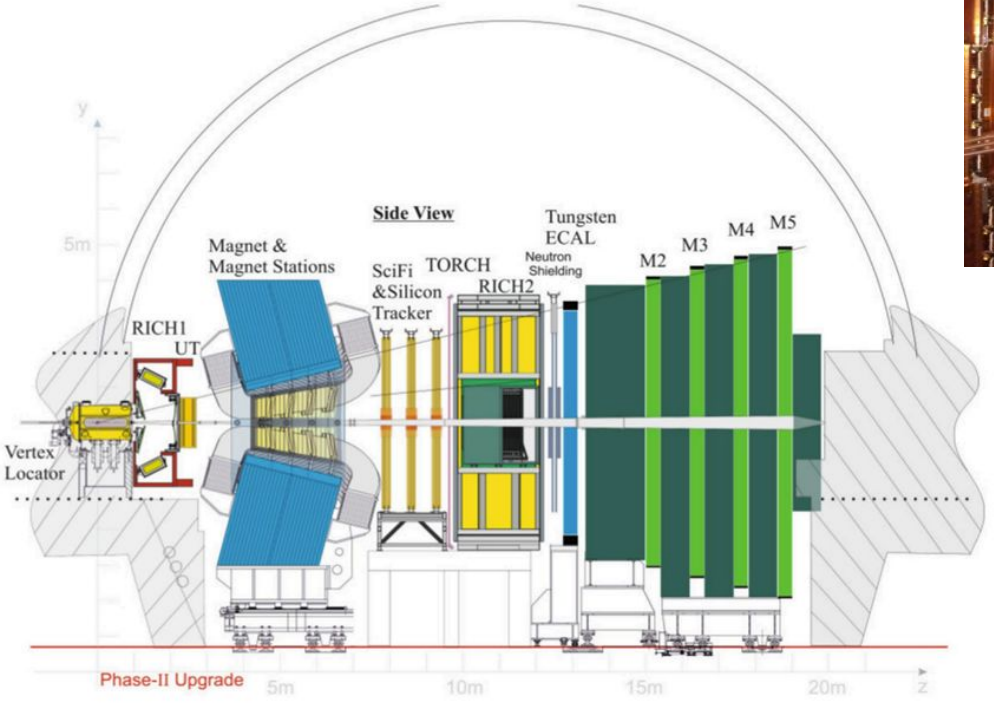
Step 7 – **Electrical cleaning** and detector closing @ CERN

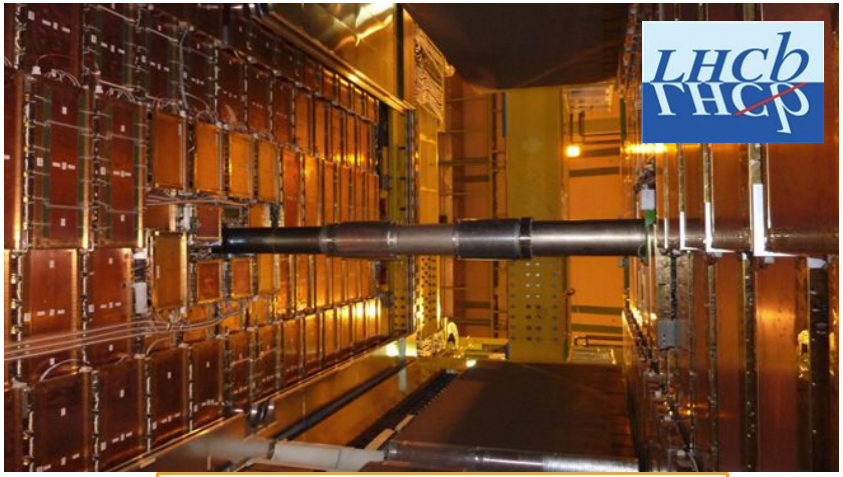


LHCb upgrade II (Run 5-6)

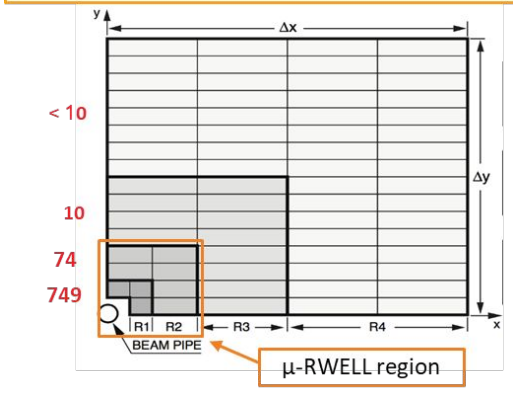
LHCb muon RUN 5-6 option: μ-RWELL ⟶ Detector requirements:

- Efficiency (4 gaps) > **99% within BX** (25 ns)
- Stability up to 1 C/cm^2 accumulated charge in 10y of operation





M2 station - max rate (kHz/cm^2)



LNF MPT Workshop - Photolithographic processes



1 – APICAL 5/50/0 base material



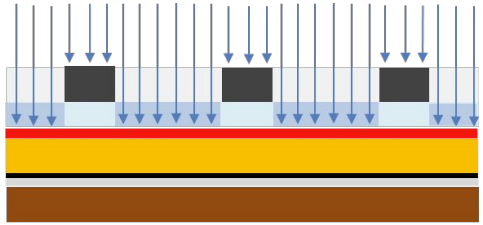
2 – DLC sputtering



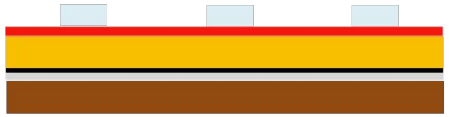
3 – DLCed base material on PCB substrate



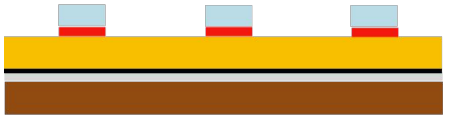
4 – Dry film lamination



5 – UV-exposure with negative mask



6 – Wet Dry-film development



7 – Cu etching



8 – Unexposed Dry film stripping



9 – PI etching

LNF MPT Workshop - Status and medium-term Schedule

- Preliminary project development: Nov 2024 – Jan 2025
- Project discussion with the LNF Director & the Head of the Research Div.: Feb 2025
- Selection of the laboratory space
- Clearing and cleaning of the premises: completed on April 15, 2025
- Identification of machinery and chemical materials: ongoing
- Meeting at ELTOS for final project definition: April 28, 2025
- 1st discussion with the Technical Division and DIV-RIC Director: May 8, 2025

- Contacts with suppliers: ongoing
- Procurement of Phase-1 equipments: within 2025 (delivery within middle 2026)
- Refurbishment of the facility building:
  - Design phase: within September 2025
  - Renovation works: to be completed by end of 2026/mid-2027
- Commissioning of the facility: second half of 2027- first half 2028